

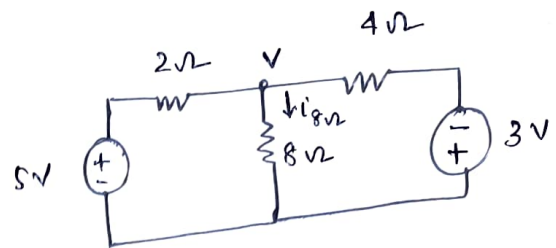
1. Ans:

Apply nodal analysis at node v

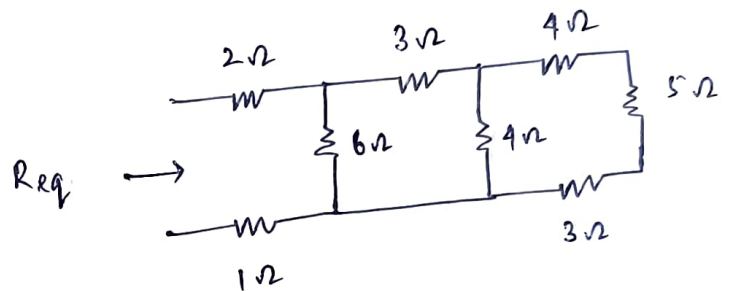
$$\frac{V-5}{2} + \frac{V}{8} + \frac{V-(-3)}{4} = 0$$

$$\Rightarrow V = 2V$$

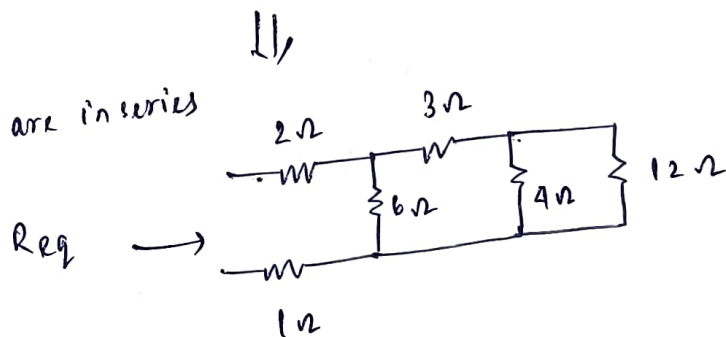
current through 8Ω resistor is $\frac{V}{8} A = \frac{2}{8} A = \boxed{0.25 A}$



2. Ans.

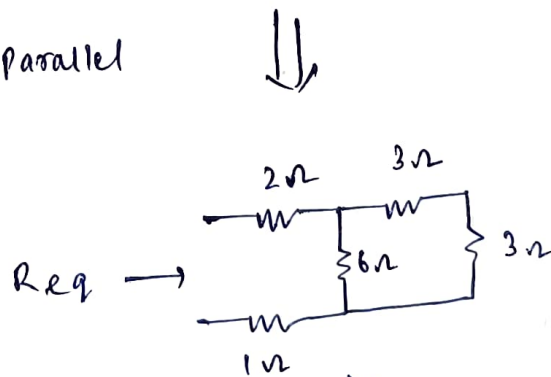


4Ω , 5Ω and 3Ω are in series
so, $R_{eq} = 12\Omega$

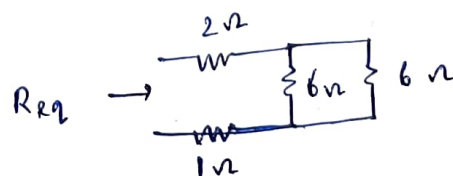


12Ω & 4Ω are in parallel

so, $R_{eq} = 3\Omega$



3Ω & 3Ω are in series



Finally

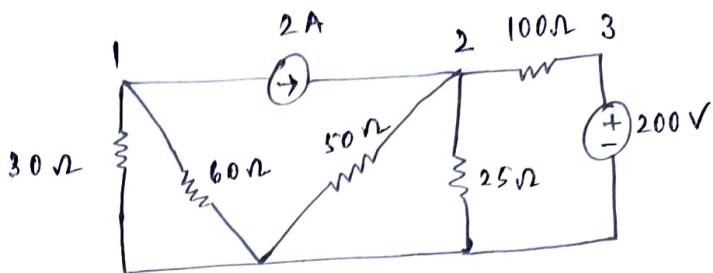
$$R_{eq} = (2+1) + (6||6) = 6\Omega$$

3. Ans.

Apply nodal analysis at node 1

$$\frac{V_1}{30} + \frac{V_1}{60} + 2 = 0$$

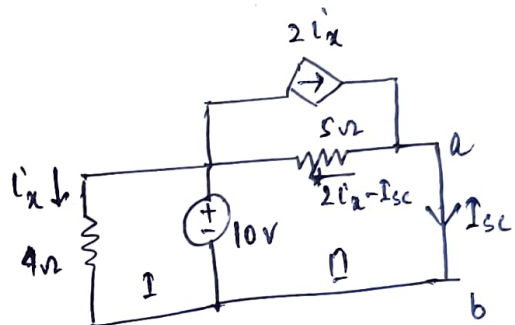
$$V_1 = -40 \text{ V}$$



4. Ans:-

To get I_{sc}

step 1:- short circuit the a-b terminal



Apply KVL in loop 1

$$10 = 5(I_{sc} - 2i_x) \quad \text{--- (1)}$$

Also, Apply KVL in loop 2

$$10 = 4i_x \quad \text{--- (2)}$$

$$\textcircled{2} \Rightarrow i_x = 2.5 \text{ A}$$

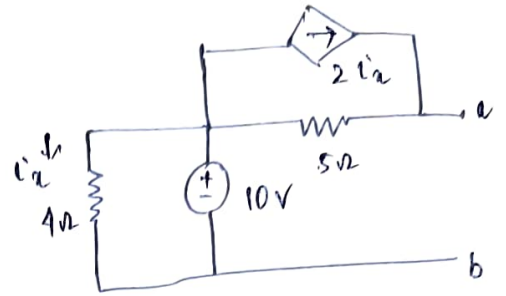
$$\textcircled{1} \Rightarrow I_{sc} = 7 \text{ A}$$

To get R_N , we need V_{th} & I_{sc}

so, to get V_{th}

apply nodal analysis at a-b terminal

$$-2i_a + \frac{V_{ab} - 10}{5} = 0 \quad \text{--- (1)}$$



Also,

$$i_a = \frac{10}{4} = 2.5 \text{ A}$$

so, (1) $\Rightarrow$ ~~i_a~~ $V_{ab} = 35 \text{ V}$

$$V_{ab} = V_{th}$$

so,

$$R_N = \frac{V_{th}}{I_{sc}} = \frac{35}{7} = 5 \Omega$$

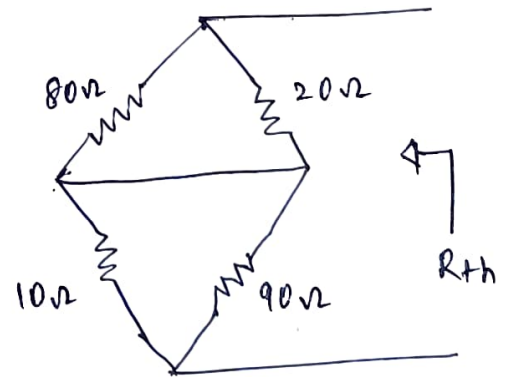
(5)

For max^m power transfer $\Rightarrow R = R_{th}$

to calculate R_{th} , we need to remove R and deactivate independent source in the circuit

so, 80Ω & 20Ω are connected in parallel

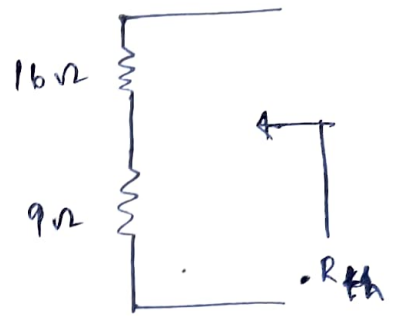
$$so, R_{eq} = \frac{80 \times 20}{80 + 20} = 16 \Omega$$



and 10Ω & 90Ω are connected in parallel

$$so, R_{eq} = \frac{10 \times 90}{10 + 90} = 9 \Omega$$

So, $R_{th} = (16 + 9) \Omega = 25 \Omega$



⑥

Given $i = \cos 4t$

$v = \sin 4t$

For Resistor $v = iR$

* Resistor provides only attenuation or amplification in current waveform for given voltage waveform

* voltage across inductor $v_L = L \frac{di}{dt}$

if we differentiate $i = \cos 4t$ with respect to t , we will get $-4 \sin 4t$.

* current in capacitor $= C \frac{dv}{dt}$

if we differentiate $v = \sin 4t$ with respect to t , we will get $4 \cos 4t$

so,

For the given question, capacitor will be correct answer.

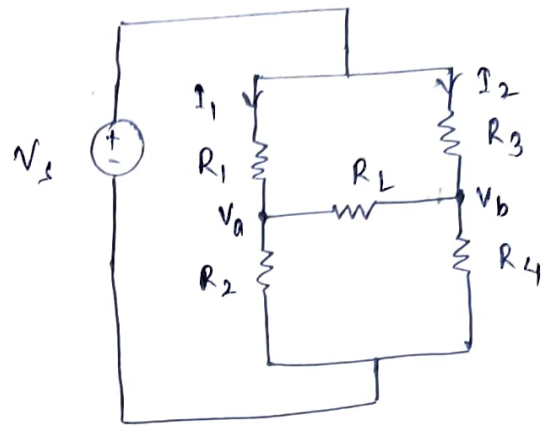
⑦ Given that no current is flowing through R_L

then

$$\boxed{V_a = V_b} \quad \text{--- (1)}$$

Also,

R_1 & R_2 are in series }
 i.e., R_3 & R_4 are in series } --- (2)



$$\text{so, } I_1 = \frac{V_s}{R_1 + R_2} \quad \text{--- } I_2 = \frac{V_s}{R_3 + R_4}$$

$$V_a = \frac{V_s \times R_2}{R_1 + R_2}, \quad V_b = \frac{V_s R_4}{R_3 + R_4}$$

$$\textcircled{1} \Rightarrow V_a = V_b$$

$$\text{so, } \frac{\cancel{V_s} R_2}{R_1 + R_2} = \frac{\cancel{V_s} R_4}{R_3 + R_4}$$

$$R_2 R_3 + \cancel{R_2 R_4} = R_1 R_4 + \cancel{R_2 R_4}$$

$$\text{so, } \boxed{R_1 R_4 = R_2 R_3}$$

8

$$E = I^2 R \times t$$

$$300 = (2)^2 \times R \times 15$$

$$\text{so, } R = 5 \Omega$$

$$\text{so, } E = (3)^2 \times (5) \times 10 = 450 \text{ J}$$

9

$$R = \frac{V^2}{P} = \frac{220^2}{100} = 484 \Omega$$

$$\text{so, } P = \frac{V^2}{R} = \frac{110 \times 110}{484} = 25 \text{ W}$$

10

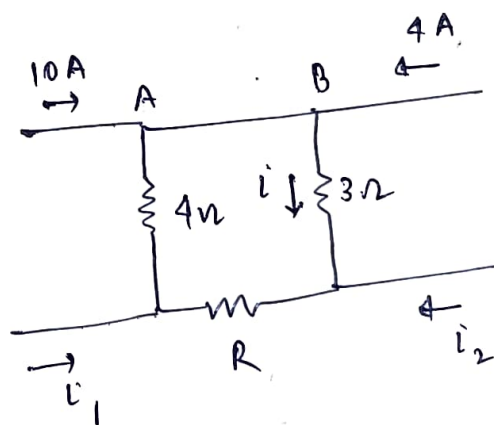
Given $R = 0$

Apply nodal analysis at node A

$$-10 + \frac{V_A}{4} - 4 + \frac{V_A}{3} = 0$$

$$\Rightarrow V_A = 24 \text{ V}$$

$$I = \frac{V_A}{3} = 8 \text{ A}$$



(ii)

Draw the circuit when $V_2 = 0$ given $i_2 = -6 \text{ A}$

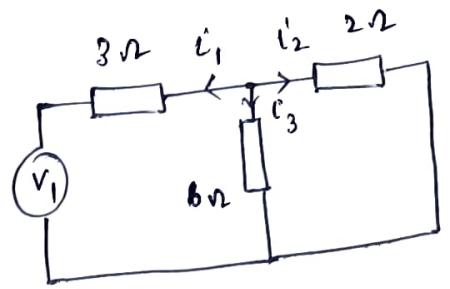
So, 6Ω & 2Ω are connected in parallel, so, current flowing

them will be inversely proportional to it

$$\frac{6}{2} = \frac{i_2}{i_3}$$

$$3 = \frac{-6}{i_3}$$

$$i_3 = -2 \text{ A}$$

Draw the circuit when $V_1 = 0$

So, given $i_1 = 6 \text{ A}$

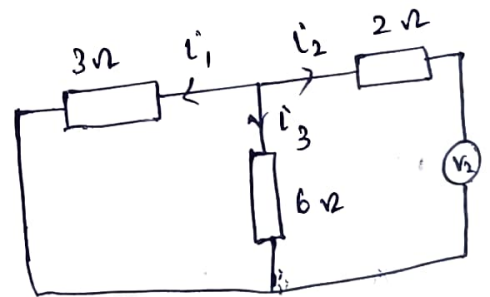
Here, 3Ω and 6Ω are in parallel

so, current flowing them will be inversely proportional to them

$$\text{so, } \frac{3}{6} = \frac{i_3}{i_1}$$

$$\frac{1}{2} = \frac{i_3}{6}$$

$$i_3 = 3 \text{ A}$$



so,

$$i_{3\text{net}} = (-2 + 3) \text{ A} = 1 \text{ A}$$

(12) Kirchhoff law states that at a junction

$$\sum i_{\text{entering}} = \sum i_{\text{leaving}}$$

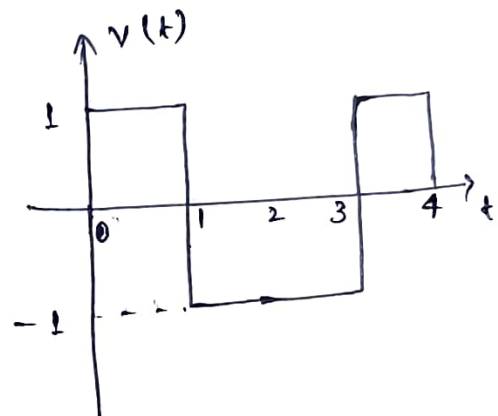
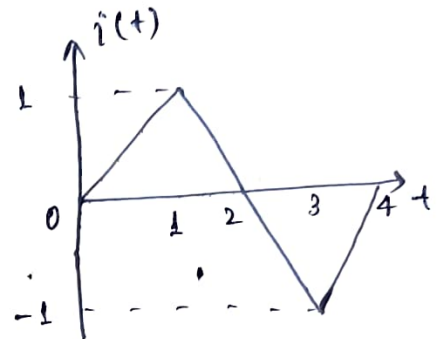
(13)

$$v = L \frac{di}{dt}; \quad i(t) = \begin{cases} t & 0 \leq t < 1 \\ 2-t & 1 \leq t < 3 \\ t-4 & 3 \leq t < 4 \end{cases}$$

Differentiate the current waveform

We get

$$v(t) = \begin{cases} 1 & 0 \leq t < 1 \\ -1 & 1 \leq t < 3 \\ 1 & 3 \leq t < 4 \end{cases}$$



(14)

$$v_L = L \frac{di}{dt}$$

$$= 5 \times \frac{d}{dt} (5t^2 + 100)$$

$$= 5 \times 10t = 50t$$

$$\boxed{v_L \Big|_{t=10s} = 500 \text{ V}}$$

(15)

$$\tau = RC$$

$$\tau = 500 \times 12 \times 10^{-6}$$

$$\boxed{\tau = 6 \text{ ms}}$$